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Starch Utilization by Ruminants: From Basics to the Bunk^{1,2}

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ABSTRACT: Starch is the major energy component of grains. Wheat contains 77% of DM as starch, corn and sorghum contain 72%, and barley and oats contain 57 to 58%. In vitro systems have provided valuable data on kinetic aspects of starch digestion. Molecular biological techniques have provided a clearer picture of the ruminal microbial milieu. Proportions of starch fermented in the rumen can be predicted satisfactorily for a variety of grains and processing methods. Compared with dry rolling, steam processing (flaking or rolling) increases ruminal digestibility of starch (percentage of intake) from 52 to 78% for sorghum, from 75 to 85% for corn, and six percentage units or less for other grains. Recent research provides new insight into pancreatic function and intestinal glucose transport systems. The capacity to digest starch in the intestine ranges from 45 to 85% of starch entering the duodenum, with that capacity

apparently limited by the supply of pancreatic amylase. There is evidence that amylase secretion may be enhanced by increasing duodenal entry of protein. Capacity for active transport of glucose across the gut wall does not seem to limit the amount of starch digested that is absorbed as glucose. For ruminants eating medium- to high-concentrate diets, about 30% of their total glucose need comes from glucose absorption, 50% from organic acid absorption (substrates for hepatic gluconeogenesis), and 20% from other sources. When glucose absorption from the gut increases, ruminants generally adjust (decrease) gluconeogenesis to meet their need; that need is directly linked to DE intake. In terms of overall ME yield, grain starch is best used when it is fermented in the rumen. However, close coordination of protein and starch supply to the duodenum may improve capture of starch in the form of glucose.

Key Words: Starch, Ruminants, Digestion, Absorption, Glucose, Metabolism

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Introduction

For over 40 years, substantial portions of U.S. grain production have been marketed through beef, sheep, or dairy products. The basic drive for feeding high-grain diets to ruminants is the price of grains relative to forages. Furthermore, grains increase the energetic density of the diet, which will optimize production in well-managed, intensive systems. Because starch is the major energy component of grains, improvements in these intensive systems will depend upon improved conversion of starch to animal product.

Research progress in starch utilization by ruminants has been reviewed periodically, with the last major effort published in 1986 (Rooney and

Pflugfelder, 1986; Ørskov, 1986; Owens et al., 1986; Theurer, 1986). The objective of this review is to present and evaluate information made available since 1986 through individual research reports as well as reviews of specific aspects of starch digestion, glucose absorption, and glucose metabolism of ruminants eating diets that contain substantial quantities of starch (e.g., Nocek and Tamminga, 1991; Croom et al., 1992, 1993; Harmon 1992, 1993; Kotarski et al., 1992; Hediger and Rhoads, 1994).

Characteristics of Grains

A review of grains used in research during the past few years showed, as expected, that starch content of wheat (77% starch) was the highest for the grains (Aimone and Wagner, 1977; Herrera-Saldana et al., 1990b), followed by 72% starch in both corn (Herrera-Saldana et al., 1990b; Zinn, 1991, 1993a,b; Larson et al., 1993) and sorghum (Herrera-Saldana et al., 1990b; Streeter et al., 1990a,b, 1991; Zinn, 1991; Poore et al., 1993), then by 57 to 58% starch in barley and oats (Herrera-Saldana et al., 1990b; Hatfield et

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al., 1993; Zinn, 1993a,b). These general means do not reflect differences among starch sources or effects of varieties, location, year, climatic conditions, and agronomic practices. Nocek and Tamminga (1991) published more detailed information on non-structural carbohydrates for many feedstuffs. The starch content of corn silage is a direct function of plant maturity and proportion of grain in the whole plant; Mahanna (1994) reported that starch content of corn silage increased from 22 to 35% as the percentage of grain in the silage increased from 32 to 50%. Direct comparison of five grains from a local supply (Herrera-Saldana et al., 1990b) showed the smallest coefficient of variation for corn (2.4%) followed by sorghum (3.7%), wheat (4.1%), barley (5.2%), and oats (7.1%). Forty-two sorghum grain samples collected by the National Grain Sorghum Producers Association contained (on average) 68% starch with a coefficient of variation of 7.1% (Preston et al., 1991). For purposes of comparison, alfalfa hay or silage contains from 2.7% to 20% starch (Mahanna, 1994; Poore et al., 1993), and protein supplements such as soybean meal, cottonseed meal, and fish meal contain from 2.5% to 27% starch (Nocek and Tamminga, 1991; Poore et al., 1993).

Figure 1 shows bacterial colonization and digestion of corn after 48 h in the rumen (McAllister, Agriculture Canada, Lethbridge, Alberta, unpublished data), and demonstrates the resistance of the pericarp to attachment and digestion (Figure 1a), relatively greater attachment and digestion of the horny endosperm (Figure 1b) and floury endosperm (Figure 1c), and finally the extensive invasion and digestion of starch granules (Figure 1d).

The structure of grain kernels (seeds) reflects their biological function: protection and transport of the embryo, complimented with an energy source to aid in germination, and early growth. The pericarp encases the embryo or germ as well as the endosperm, which contains most of the starch (Kotarski et al., 1992). Within the endosperm is the aleurone layer, which contains essential enzymes and enzyme inhibitors. Beneath the aleurone layer are the peripheral and corneous endosperm, which contain starch granules embedded in a protein-rich matrix. The scanning electron micrographs of Rooney and Pflugfelder (1986), Chandrashekar and Kirleis (1988), and McAllister et al. (1990b) provide good visualization of those structures, as well as visualization of microbial invasion and digestion of the starch. This type of micrograph shows that "microbial attack" is not an exaggerated metaphor for the havoc wreaked by ruminal bacteria. Beneath all these layers is the floury endosperm, which has the highest concentration of starch granules that are not embedded in a protein matrix. Starch in the floury endosperm is the most susceptible to external forces such as digestion or grain processing. Starch granules are composed

mainly of amylopectin (α 1-4 and α 1-6 linkages) and amylose (α 1-4 linkages). Proportions of the two polysaccharides vary among species and varieties, with amylose contributing from 0 to about 20% of the total (Rooney and Pflugfelder, 1986; Kotarski et al., 1992). Also present are relatively small amounts of pectins and sugars, which, with starch, compose non-structural carbohydrates (Nocek and Tamminga 1991).

Differences in relative abundance of endosperm layers are the basis for descriptors such as vitreous, flinty, waxy, nonwaxy, and opaque, which are used to categorize grains and varieties. These structural differences also are the basis for reported differences in in vitro or in vivo digestion among grain sources and subsequent differences in animal performance attributable to grain species or varieties (Rooney and Pflugfelder, 1986; McAllister et al., 1990b; Streeter et al., 1990a,b, 1991; Kotarski et al., 1992; Wester et al., 1992). In general, waxy genotypes contain only amylopectin, swell faster in heated water, and have faster rates of in vitro or in vivo digestion than heterogenous or nonwaxy genotypes.

Ruminal Digestion of Starch

Although protozoa and fungi participate in ruminal digestive processes, the bulk of the fermentation is performed by ruminal bacteria. A key initial step in bacterial digestion of feed particles is attachment; approximately three-fourths of fiber, protein, and starch digestion is accomplished by bacteria that are loosely or tightly attached to feed particles (McAllister et al., 1994). Kotarski et al. (1992) identified 15 strains of amylolytic bacteria and characterized eight amylolytic enzymes produced by those bacteria. At least some of these bacteria adhere to and colonize grain particles in the rumen and produce endo- and exo-enzymes that hydrolyze the α 1-4 and α 1-6 bonds of amylose and amylopectin. However, not all bacteria are equipped with a complete array of digestive enzymes; therefore, maximal digestion of starch to monosaccharides requires integration among bacterial species. Coculture of *Streptococcus bovis*, *Butyrivibrio fibrisolvens*, *Bacteriodes ruminicola*, and *Selenomonas ruminantium* demonstrated the importance of cross-feeding among bacterial species in attaining greatest bacterial growth rates and complete digestion of starch (Cotta, 1992). Ruminal protozoa play a role in the process by ingesting and digesting starch granules (with or without bacteria attached). Rate and extent of ruminal starch digestion were greater when protozoa were eliminated from the rumen of sheep fed high-moisture corn and dry-rolled sorghum (Mendoza et al., 1993). The hyphae of fungi may play an important role in bacterial attachment by creating lesions in the surface of plant tissue (McAllister et al., 1994).

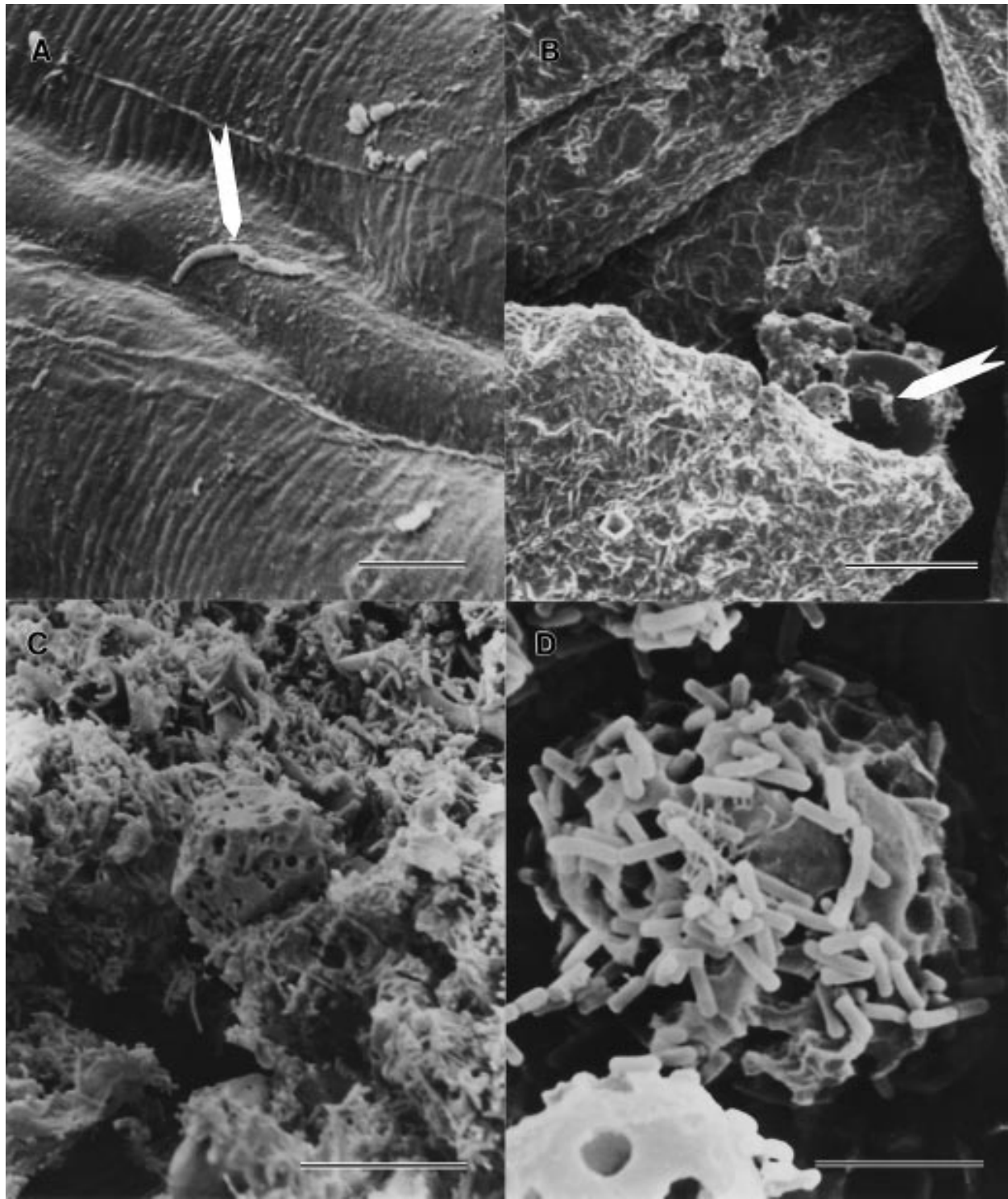


Figure 1. Scanning electron microscopy of corn after 48 h of incubation in the rumen (McAllister, unpublished). Panel A: Scanning electron micrograph of the pericarp of corn. Note that this structure is colonized by a limited number of bacteria and that there is no evidence of digestion. Bar = 5 μm . The white arrow indicates bacteria. Panel B. Scanning electron micrograph of the horny endosperm of corn. Note that this region is colonized by a limited number of bacteria. Bar = 20 μm . The white arrow indicates bacteria. Panel C. Scanning electron micrograph of the floury endosperm of corn. Note that the surface of this region is covered by a thick biofilm of bacteria. Bar = 5 μm . Panel D. Scanning electron micrograph of a colonized starch granule within the floury endosperm of corn. The starch granule is colonized by numerous rod-shaped bacteria and digestive pits on the surface of the granule are readily apparent. Bar = 5 μm .

Whole grain with an intact pericarp is largely or entirely resistant to digestion by ruminants because whole kernels are resistant to bacterial attachment (Figure 1; McAllister et al., 1994; Beauchemin et al., 1994). Conversely, grain is processed by application of various combinations of heat, moisture, time, and mechanical action. These processing treatments increase digestibility of the starch by providing opportunity for bacterial attachment to starch granules. Rooney and Pflugfelder (1986) described the chemical and physical changes that popular methods of grain processing elicit, e.g., gelatinization, retrogradation, and dextrinization of the grain starch. Ruminants "process" starch by chewing, thereby breaking the pericarp.

Rate and extent of starch digestion in the rumen are determined by intricate interrelations among several factors, including source of dietary starch, diet composition, amount of feed consumed per unit time, mechanical alterations (grain processing, chewing), chemical alterations (degree of hydration, gelatinization), and degree of adaptation of ruminal microbiota to the diet. Three approaches to control rate and extent of starch digestion that have received the most attention are management of feed consumption, grain processing, and feed additives. Owens and Goetsch (1986) provided a comprehensive review of the effects of these approaches on key ruminal kinetic variables such as sizes of ruminal liquid and feed particle pools, chemical properties of those pools (pH, osmolarity), rates of liquid passage, rate of feed particle digestion, rate of feed particle passage (ruminal escape), rate of dilution of ruminal medium, and outflow of microbial products (microbial protein). In general (over)simplification, rate of feed particle digestion in the rumen is directly proportional to extent of digestion, and increased rate of feed particle passage is directly proportional to increased feed consumption. Therefore, altered feed consumption by management technique (Loerch, 1990; Pritchard, 1995) or by feed additives such as ionophores (Zinn, 1987; Burrin et al., 1988; Stock et al., 1990; Zinn and Borques, 1993) may affect how much starch is digested in the rumen. Similarly, additives or treatments such as dietary fat (Brandt, 1992) or treatment with formaldehyde or phenolics (Fluharty and Loerch, 1989; Mahmoudzadeh et al., 1989; McAllister et al., 1990a; Oke et al., 1991; McAllister et al., 1992a,b; Castlebury and Preston, 1993) are designed to slow or limit ruminal starch digestion; however, these studies have not shown consistent results in terms of site and extent of starch digestion or in terms of animal performance.

Evaluation of sorghum hybrids has provided the best available example of the relationship between genotype and starch digestion. Sorghum hybrids vary in content of starch and protein, in structure of endosperm layers, in *in vitro* digestibility, and in their use by cattle for weight gain (Streeter et al., 1990a,b, 1991; Kotarski et al., 1992; Wester et al., 1992).

Results from these studies show waxy varieties have greater rates and extents of *in vitro* starch digestion than nonwaxy varieties. Nonwaxy varieties with a floury starch texture in their endosperm have faster *in vitro* starch disappearance rates than varieties with intermediate or vitreous endosperms. Bird-resistant varieties have slower rates of *in vitro* digestion than varieties not classed as bird-resistant. The preponderance of data indicates no strong correlation between protein content and rate of *in vitro* starch digestion (Camgampang and Kirleis, 1984; Wester et al., 1992). Rate and efficiency of weight gain by cattle or sheep do not always respond as clearly to different varieties as might be predicted by *in vitro* rankings, but in general waxy varieties (or heterogenous genotypes expressing waxy genes) have more efficient conversion of feed to weight gain.

Reviews by Owens et al. (1986) and Theurer (1986) provide comprehensive summaries of effects of grain source and grain processing methods on site and extent of digestibility of starches; data published since those reviews confirm and elaborate those summaries. Table 1 summarizes studies with beef and dairy cattle published since 1986. The number of observations (animals) per treatment ranged from three to six, with four the prevalent number. None of the studies included unprocessed grains; usual treatments were processing methods, often compared with dry-rolled corn as a control or baseline treatment. For purposes of comparison, Owens et al. (1986) reported ruminal digestibility of whole corn to be 58.9% and total tract digestibility to be 91.7%. Standard deviations were calculated for means of treatment means when there were more than three studies of a particular grain and processing method. Of course, variation among treatment means reflects differences in processing technique, grain sources, and experimental procedures. Regression of ruminal, post-ruminal or total tract digestibility of dry-rolled corn, dry-rolled sorghum, and steam-flaked sorghum as a function of daily starch intake showed no strong correlation between intake and digestibility (data not shown). That is consistent with results of a study designed to test intake effects (Zinn, 1990b).

Compared with dry rolling, steam flaking increased starch digestibility for corn and sorghum; this is apparent in the means in Table 1 as well as in results of studies designed to test processing methods. Furthermore, steam flaking versus dry rolling corn decreased the variation associated with means, suggesting that gelatinization and other alterations caused by that type of processing improved the uniformity of response. Steam rolling increased the starch digestibility in barley but had little effect on wheat or oats. Steam flaking increased the digestibility of starch in the rumen more for sorghum (19 percentage units) than for corn (13 percentage units). Preston et al. (1993) related *in vitro* enzymatic digestibility of sorghum grain starch to the degree of

Table 1. Digestibility coefficients for starch published during the past nine years

Digestibility							
Grain	Processing ^a	Starch intake, kg/d	Rumen	Postrumen		Total tract, %	Citations ^b
			% Intake	% Intake	% Entering		
Corn	DR	2.06 ± 1.08	76.2 ± 7.9	16.2 ± 6.7	68.9 ± 18.4	92.2 ± 3.0	1–8
	SF	2.20 ± .52	84.8 ± 4.1	14.1 ± 3.7	92.6 ± 4.1	98.9 ± 0.8	5–13
	SR	6.91	72.1	19.0	68.2	91.2	14
	HM	3.89	89.9	6.3	67.8	95.3	15
	G	10.65	49.5	44.0	86.5	93.5	16
Sorghum	DR	4.81 ± 1.49	59.8 ± 12.0	26.1 ± 11.4	62.5 ± 16.2	87.2 ± 5.4	1, 4, 15, 17–19
	SF	4.78	78.4	19.6	89.9	98.0	10, 14, 18
	HMG	3.64	73.2	19.6	46.1	92.8	2, 20, 21
	G	3.81	70.0	15.4	51.0	91.0	21
Barley	DR	4.09 ± 1.74	80.7 ± 3.9	13.7 ± 3.8	75.2 ± 5.2	94.3 ± 2.9	16, 22–24
	SR	4.53	84.6	13.6	88.0	98.2	8
Wheat	DR	2.94	88.3	9.9	85.4	98.2	20, 24
	SR	2.87	88.1	10.0	88.2	98.6	24
Oats	DR	1.53	92.7	5.6	76.3	98.3	12
	SR	1.49	94.0	4.5	78.8	98.8	12

^aDR = dry-rolled; SF = steam-flaked; HM = high moisture; G = ground; SR = steam-rolled.

^bCitations: 1. Spicer et al. (1986); 2. Streeter et al. (1989); 3. Streeter and Mathis (1995); 4. Streeter et al. (1990b); 5. Zinn (1987); 6. Zinn (1988); 7. Zinn (1990a); 8. Zinn et al. (1995); 9. Zinn (1990b); 10. Zinn (1991); 11. Zinn (1993a); 12. Zinn (1993b); 13. Zinn and Borques (1993); 14. Oliveira et al. (1995); 15. Stock et al. (1987a); 16. McCarthy et al. (1989); 17. Herrera-Saldana et al. (1990a); 18. Poore et al. (1993); 19. Streeter et al. (1991); 20. Axe et al. (1987); 21. Hill et al. (1991); 22. McNiven et al. (1995); 23. McAllister et al. (1992b); 24. Zinn (1994).

gelatinization of the starch, and they found that starch digestibility (availability) was positively related to percentage of starch that was gelatinized; however, availability did not improve substantially when more than 60% of the starch was gelatinized. This in vitro relationship fits well with observed feedlot performance in response to flake density of steam-flaked sorghum. Starch availability and efficiency of gain both increased as starch availability reached 70% of total starch (Xiong et al. 1991).

Because there is no strong relationship between starch intake and ruminal digestibility, there is no apparent limit to the capacity of the rumen to digest starch. However, almost all the adversities associated with feeding high-grain diets (e.g., bloat, acidosis, founder, abscessed livers) are the result of excessively rapid fermentation of starch to organic acids. It follows that most feed additives, feed treatments, and management techniques designed to ameliorate these adversities focus on ways to slow the fermentation rate or neutralize the acids produced. Similarly, the main goal of research on grain-processing techniques has been to increase digestibility of grain starch yet avoid "too much of a good thing" by making starch too available for microbial attack.

Use of ruminal buffers or neutralizers closely followed identification of acidosis as the main cause of digestive upsets associated with consumption of high starch diets. Today, compounds such as sodium bicarbonate, calcium carbonate, or magnesium oxide are added to ameliorate those digestive upsets or to maintain milk fat percentage when high grain diets are fed to lactating cows. Recent work changes the

perception of how these compounds work. In his review, Erdman (1988) could find no relationship between ruminal fluid pH and the percentage of milk fat in dairy cows. Additionally, he found no relationship between dietary sodium bicarbonate or magnesium oxide and blood or urine pH, blood pCO₂, or blood bicarbonate concentration. Aslam et al. (1991) tried unsuccessfully to show that sodium bicarbonate delivered to the rumen up to 6 h after feeding would alter ruminal concentration of organic acids and therefore alter the milk fat percentage. Adding commercially available buffers to dairy cow diets increased the milk fat percentage sometimes, but had no effects on ruminal fluid pH or organic acid concentrations (Xu et al., 1994). Zinn and Borques (1993) found no effect of dietary sodium bicarbonate on ruminal fluid characteristics, site or extent of starch digestion, or feedlot performance of steers fed high grain diets. Lack of relationships between feeding buffers and metabolic or physiological variables of acid-base status does not support the idea that these compounds function as ruminal or metabolic buffers. After they could not establish any relationship between a "buffer value index" in diets containing from 50 to 70% grain and milk yield, milk fat, or milk protein, Miller et al. (1993) decided that feedstuffs contribute about 15% of the total buffering capacity available to cows. After their review of literature and biochemical calculations, Russell and Chow (1993) concluded that physiologically it was unlikely that dietary buffers could have much of an effect on ruminal fluid pH relative to the predominant effect of transfer of CO₂ from blood. They proposed that the mechanism of action of sodium bicarbonate is to

increase water consumption, increase dilution of ruminal fluid, and subsequently increase the amount of starch that escapes fermentation. Wiedmeier et al. (1987) increased salivary output of cows with pharmacological agents and fed them either sodium chloride or sodium bicarbonate. All treatments increased the liquid dilution rate in the rumen and increased rate of particulate passage from the rumen. Increased salivary output decreased ruminal fluid volume but did not change pH; both sodium salts increased ruminal liquid volume and pH. The authors concluded that sodium chloride and sodium bicarbonate were equally effective in increasing ruminal fluid pH. Intramuscular injection of slaframine increased salivation, ruminal fluid pH, rates of liquid and particulate passage from the rumen, microbial protein production, and ruminal digestion of starch in steers (Froetschel et al., 1995).

Synchronization of Ruminal Energy and Nitrogen Availability

The relatively high rate of absorption of ammonia by ruminants (Huntington, 1990) suggests that energy availability, or lack of synchrony between energy and nitrogen supplies, limits the use of available nitrogen by ruminal microorganisms. This in turn limits efficient use of dietary nitrogen for productive purposes. Synchronization of starch and nitrogen supply to the rumen reduced absorption of ammonia and increased nitrogen retention in steers (Taniguchi et al., 1995). Synchronization of ruminal fermentability of starch and protein increased nitrogen retention of growing lambs as a percentage of nitrogen intake (Matras et al., 1991). Readily available energy from starch fermentation is conducive to increased ruminal outflow of microbial protein in cattle (Spicer et al., 1986; Zinn, 1988, 1993a,b; Streeter et al., 1989; Poore et al., 1993). For example, ruminal outflow of bacterial protein increased from 2.2 to 2.8 kg/d in dairy cows fed steam-flaked vs dry-rolled sorghum (Poore et al., 1993). Other studies with lactating cows (Herrera-Saldana and Huber, 1989; Herrera-Saldana et al., 1990a; Aldrich et al., 1993) confirm that synchronizing the ruminal fermentability of starch and protein (nitrogen) sources increases outflow of bacterial protein from the rumen. Two studies with lactating cows fed corn or barley as starch sources did not show clear response to ruminal synchronization (McCarthy et al., 1989; Khorasani et al., 1994). In one of those studies, the ability to detect responses may have been compromised by differences in intake between diets containing ground, shelled corn and steam-rolled barley (McCarthy et al., 1989).

Intestinal Starch Digestion

Enzymatic digestion of starch in the small intestine of ruminants proceeds much as it does in other

species. The pancreas secretes α -amylase (EC 3.2.1.1), which hydrolyzes amylose and amylopectin into limit dextrins (principally from hydrolysis of amylopectin) and linear oligosaccharides of two to three glucose units (Gray, 1992; Harmon, 1993). The process is completed by surface oligosaccharidases that are located on the brush border membrane of intestinal microvilli. However, unlike nonruminants, ruminants do not have measurable sucrase activity and therefore depend on maltase and isomaltase activity to produce glucose units for absorption (Harmon, 1992).

The advent of feeding high-grain diets promoted interest in determining the capacity of ruminants to digest starch in the intestine, because absorption and metabolism of glucose seem to be more efficient energetically than fermentation and absorption of organic acids (Owens et al., 1986). Harmon (1993) concluded in his review that secretion and production of starch digestion enzymes in the intestine respond more to the amount of energy consumed by the animal than to the amount of dietary starch. Croom et al. (1992) concluded that by virtue of being pregastric fermenters, ruminants have a relatively constant flow of digesta to the duodenum, which precludes implementation of some of the neuro-endocrine control of pancreatic secretion observed in other species.

On average, 5 to 20% of starch consumed is digested postruminally, with most of that digestion in the small intestine (Streeter et al., 1989, 1991; Hill et al., 1991; Zinn, 1991). In Table 1, there are three exceptions to this range. First, the data for ground corn in Table 1 are from the lactating cows of McCarthy et al. (1989). To my knowledge, those cows had the highest starch intakes ever reported in this type of intensive study, which likely promoted very rapid rates of liquid and particulate passage from the rumen. Increased rate of passage exerted a "washout" of ground corn from the rumen and presented over 4 kg of starch daily to be potentially digested postruminally. Second, the studies with dry-rolled sorghum show that the corneous endosperm of that grain is resistant to that processing (Theurer, 1986). Third, compared with dry-rolled sorghum, time and moisture effects associated with high moisture, ground, ensiled sorghum had a positive effect on ruminal digestibility, but a negative effect on postruminal digestibility of starch. Total tract digestibility of high moisture sorghum was greater than that of dry-rolled sorghum but less than that of steam-flaked sorghum (Stock et al., 1987b).

In his review, Harmon (1992) reported that intestinal digestibilities of starch in steers or heifers (220 to 395 kg BW) ranged from 17.3 to 84.9% of starch entering the duodenum or from 122 to 636 g/d. Dietary starch intakes (from corn, barley, or sorghum) ranged from 2.0 to 4.6 kg/d. Hill et al. (1991) found a linear fit of 45% disappearance in the small intestine of steers when 430 to 1,200 g of sorghum starch entered the duodenum. Kreikemeier et al.

(1991) infused cornstarch in the abomasum of steers and recorded a decrease in disappearance from 86 to 55% as the amount of starch infused increased from 480 to 1,440 g/d. Owens et al. (1986) found a similar range in their review of literature (88 to 45% digestibility of corn or sorghum starch in the small intestine of cattle). The slope of a linear regression through their collection of data indicated 55% of cornstarch entering the small intestine disappeared there. These studies confirmed earlier work that identified lack of adequate pancreatic amylase activity as the primary reason that there is not 100% digestion of starch in the small intestine; they also implied that pancreatic amylase secretion continued to increase with an increase in energy intake.

Nutrient Absorption and Release into the Bloodstream

The two main routes for transfer of glucose from the lumen of the small intestine to the bloodstream are active transport and paracellular diffusion in concert with absorption of water ("solvent drag"). Glucose transporters have been characterized for a variety of species, including ruminants (Hediger and Rhoads, 1994; Shirazi-Beechey et al., 1995; Bird et al., 1996). Characterization of the sodium-glucose transporter located in the intestine (**SGLT1**) includes information on chromosomal location of genes responsible for its synthesis, its primary structure and conformation in the brush border membrane, and the mechanisms by which it transports one molecule of glucose and two Na^+ in each of its cycles. Estimates of the capacity of SGLT1 transporters range from 50 to over 200 cycles per second (Hediger and Rhoads, 1994, Ferraris et al., 1989). There is a great deal of homology (80% or greater) of SGLT1 structure among species, which facilitates extrapolation of data from other species to ruminants. The density of SGLT1 transporters is a function of the surface area of the microvilli, which in turn is a function of number and conformation of intestinal villi and location within the small intestine. Comparison of intestinal surface area among a variety of species found that surface area was proportional to $\text{BW}^{.70}$ (Ferraris et al., 1989). The highest density of transporters is in the proximal portion of the intestine, with fewer in the middle portion and fewest in distal portions of the intestine (Ferraris et al., 1989). Estimates of SGLT1 transporter numbers per cell generally range from 10^5 to 10^7 per enterocyte. The SGLT1 transporters are synthesized while enterocytes reside in the villus crypts, then mature and become active as the enterocytes mature and migrate to the tips of the villi (Shirazi-Beechey et al., 1995). Research with ruminants (Shirazi-Beechey et al., 1995) coincides with results from other species (Diamond and Karasov, 1984) that showed rapid

adaptation to changes in increased dietary supply of starch. The capacity to actively transport glucose may increase twofold over a period of 2 to 4 d.

Simulation of Starch Digestion and Glucose Uptake from the Small Intestine

I used the basic information outlined above to derive mathematical fits to the data of Kreikemeier et al. (1991). Those data provide information on digesta flow and concentrations of starch and glucose, intestinal disappearance, and net portal absorption of glucose in steers eating alfalfa hay alone or eating alfalfa hay plus receiving three levels of abomasal glucose or starch. The small intestine was arbitrarily chosen to be 25 m long and to have a volume of 3.02 L (Kreikemeier et al., 1990). I distributed transport activity in the ratio 5.4:3.75:1 proximal:middle:distal in the intestine (Ferraris et al., 1989), then fitted those values to a quadratic equation with length of intestine as the independent variable. Water absorption was fixed at .8 times volume per day, and paracellular diffusion of glucose was extrapolated from parameters from rats (Pappenheimer and Reiss, 1987). The condensed version of the equation for paracellular diffusion (grams per day) is .743 times millimolar luminal glucose concentration. The intestine was mathematically divided into 25 1-m sections to iterate equations. After each iteration, fluid flow and glucose concentrations were adjusted for glucose uptake and water absorption. These adjusted values were used to determine transport and paracellular diffusion in the next section of the intestine.

If maximal active transport was set at 757 g/d and K_m set at 5 mM luminal glucose concentration, the glucose data of Kreikemeier et al., (1991) could be duplicated with reasonable agreement (Table 2). Maximal transport was calculated to be 60.7 g/d for the first meter of the intestine and declined quadratically to .9 g/d for the last meter. Average maximal transport was 30.3 g/d. Simulated transport (Table 2) approached maximum at 1,920 g/d of glucose infusion; maximal luminal glucose concentration in that simulation was 103 mM. At 1,920 g/d glucose infusion, the capacity of the intestine to transport or passively diffuse glucose was exceeded, and ileal glucose concentration increased appreciably. Diffusion was a small portion of total uptake but increased as the amount of glucose infused increased and luminal glucose concentrations increased.

Estimated starch digestion then was added to calculated glucose transport and diffusion by fitting the Michaelis-Menten equation to starch digested as a function of length of the intestine (Table 2). I was not able to simulate extent of starch disappearance as well as the observed luminal concentrations reported by Kreikemeier et al. (1991) until maximal rates were

Table 2. Predicted and observed^a intestinal disappearance, ileal concentrations of glucose and starch, and predicted uptake of glucose in steers

Item	Entering the duodenum, g/d			
	480	960	1,440	1,920
Glucose infused				
Disappeared, observed	466	830	1,026	
Disappeared, predicted	479	797	970	1,123
Active transport	448	664	705	720
Paracellular diffusion	31	133	265	403
Starch infused				
Digested, observed	411	644	793	
Digested, predicted	417	637	802	943
Active transport	397	551	615	645
Paracellular diffusion	20	48	81	114
Ileal glucose, mM				
Glucose infused				
Observed	1.11	24.56	61.24	
Predicted	.2	21.00	60.00	103
Starch infused				
Observed	2.49	4.37	5.64	
Predicted	1.05	4.92	13.7	23.8

^aData of Kreikemeier et al. (1991).

placed slightly higher than observed disappearance for each level of starch infusion, or until K_m (distance from the front of the intestine that marked 50% completion of digestion) was close to the front of the intestine (1.5 m). This pattern of enzymatic starch digestion fits well with the observation of Krebhel et al. (1996), who reported that the bulk of starch digestion in steers was in the proximal half of the small intestine. Increased maximal capacity to digest starch implies increased pancreatic secretion of amylase and/or increased activity of intestinal oligosaccharidases in response to increased starch infusion. Amylase secretion by the pancreas or intestinal oligosaccharidase activity may increase in correlation with, but not necessarily in direct response to, increased consumption of grain (Harmon, 1993). Walker and Harmon (1995) recently reported no increase in pancreatic secretion or concentration of amylase in steers in response to abomasal infusion of partially hydrolyzed starch. Increased potency of amylase secreted would require some type of positive interaction between the enzyme and its substrate, or protection of the enzyme from degradation by proteolytic enzymes that also are secreted by the pancreas.

The simulation oversimplifies the *in vivo* situation by assuming steady-state digesta flow and intestinal volume over 24 h and by assuming water absorption and paracellular diffusion are the same in each section of the intestine. Fluid additions such as pancreatic secretions are not considered. Starch digestion to glucose is not separated into products of amylase and oligosaccharidase activity. Fermentation of glucose is not considered, although the possibility for such fermentation exists (Nicoletti et al., 1984). Furthermore, the simulation assumes basal activity of SGLT1

transporters, or no adaptation to starch entry into the duodenum. Steers in the reference data set (Kreikemeier et al., 1991) were not adapted to starch. The pattern of starch digestion is compatible with rapid, early activity of amylase in the duodenum followed by peak intestinal oligosaccharidase activity in the jejunum (Harmon, 1992). The pattern of glucose absorption along the intestine is similar to the pattern for other nutrients in sheep, including minerals, lipids, and protein (Sklan and Hurwitz, 1985; Sklan and Halevy, 1985; Sklan et al., 1985). It is consistent with greater disappearance and net absorption of glucose from the proximal vs the distal half of the small intestine of steers (Krehbiel et al., 1996). The K_m for the SGLT1 transporter is below the range (6 to 23 mM) of *in vivo* K_m for other mammals (Ferraris et al., 1990); however, the K_m is higher than the .8 mM reported from *in vitro* studies of SGLT1 activity (Hediger and Rhoads, 1994) and does not solve the enigma of the actual concentrations in the unstirred water layer that surrounds the microvilli of the intestine. The relationship of active transport to paracellular diffusion is similar to that derived by Pappenheimer and Reiss (1987) from their data from rat intestine. They showed equal contribution of the two avenues of uptake when luminal glucose concentration is approximately 220 mM, versus 180 mM in unadapted steers (Figure 2). If maximal glucose transport is doubled, simulating an adapted steer, then transport and diffusion approach 50% each of total transfer at luminal glucose concentration of 320 mM (data not shown). Maximal luminal glucose concentration in all simulations was 103 mM when 1,920 g/d of glucose was infused into unadapted steers. Figure 2 shows that luminal glucose concentrations

for 757 or 1,514 g/d maximal SGLT1 activity did not exceed 35 mM. These concentrations agree well with those for other species (Ferraris et al., 1990), with the possible exception of a rat eating a high sucrose diet. The relatively small contribution of paracellular diffusion in the simulations is supported by net portal absorption of small but detectable amounts of 2-deoxyglucose by steers receiving duodenal infusions of that glucose analog (Krehbiel et al., 1996); 2-deoxyglucose is not transported by SGLT1.

As discussed previously, it is reasonable to expect a rapid increase in glucose transport activity as ruminants adapt to starch entering the duodenum. In addition to evidence of increased SGLT1 transporters in the intestine of sheep in response to duodenal supply of glucose or starch (Shirazi-Beechey et al., 1995), Bauer et al. (1995) found increased transport of glucose after a 3-d adaptation to an infusion of partially hydrolyzed starch into the abomasum of steers and wethers. They also provided evidence of the role of SGLT1 in glucose transport, because concomitant infusion of phlorizin (a competitive inhibitor of SGLT1 activity) decreased glucose transport in both unadapted and adapted animals. The effect of increased SGLT1 transport is shown in Figure 3. The first panel shows the response to 1,440 g/d of starch entering the duodenum and 802 g/d starch digested in the small intestine. The other panels show the effect of the same starch entry and digestive capacity, but with increased maximal SGLT1 transport. Figure 3 shows that transport activity is the primary limitation to the amount of glucose absorbed from starch digestion in the unadapted steer. The amount of starch digested exceeds the maximal transport capacity, so increased starch digestion in this situation would not result in greatly increased uptake of glucose. In the proximal third of the intestine, simulated luminal glucose concentrations (Figure 2) are five times the K_m of the transporter in the simulation (5 mM), so paracellular diffusion would transfer more glucose from the lumen if capacity to digest starch were increased beyond the 802 g/d in the simulation. However, when maximal transport capacity is increased, starch digestion becomes the primary limitation. The SGLT1 transport can essentially account for glucose from starch digestion in the first two thirds or one third of the intestine when transport capacity is increased 1.5 or 2.0 times, respectively (Figure 3). In these simulations, luminal glucose concentrations decrease below the simulated K_m of the SGLT1 transporter (Figure 2), which then limits SGLT1 activity. Paracellular diffusion (represented by the area between lines for SGLT1 transport and total uptake) accounts for 11.2% of total uptake when maximal transport capacity is 757 g/d, but decreases to 5% or less when transport activity is increased. Therefore, the simulation seems to concur with increased transporter activity in response to luminal glucose (Shirazi-Beechey et al., 1995) and the conclusion of Owens et al. (1986) that digestive

capacity is the primary limitation to maximal absorption of glucose from starch digestion. By using the percentages in Table 1, one can create an average feedlot steer consuming 6 kg of starch daily and digesting postruminally 1.03 kg of dry-rolled corn, 1.62 kg of dry-rolled sorghum, .858 kg of dry-rolled barley, .600 kg of dry-rolled or steam-rolled wheat or .336 kg of dry-rolled oats. Intestinal starch digestion of 802 g/d in the simulation (Figure 3) would allow approximately one half of the maximal postruminal supply of starch (from dry-rolled sorghum) to be digested in the small intestine and absorbed as

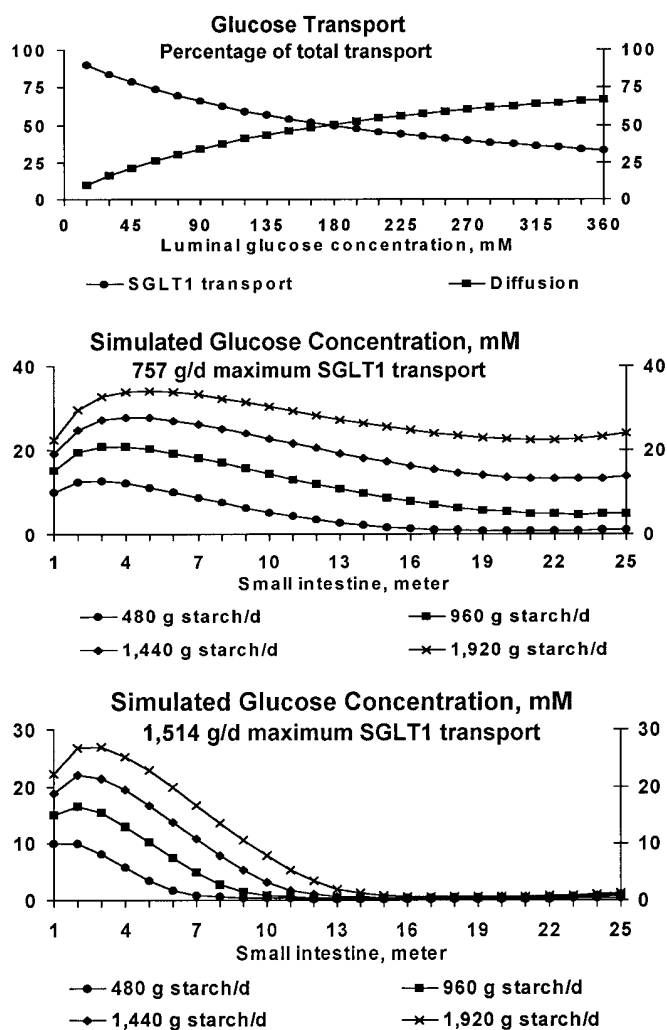


Figure 2. The top panel shows the relative contribution of SGLT1 transport and paracellular diffusion in the small intestine of a steer as putative luminal glucose concentrations vary from 0 to 360 mM. Paracellular diffusion does not exceed SGLT1 transport until glucose concentrations exceed 180 mM. The center panel shows the simulated glucose concentrations along the small intestine at four levels of starch entry into the intestine. The bottom panel shows the similar glucose concentrations as in the center panel, except maximal SGLT1 transport is now 1,514 g/d.

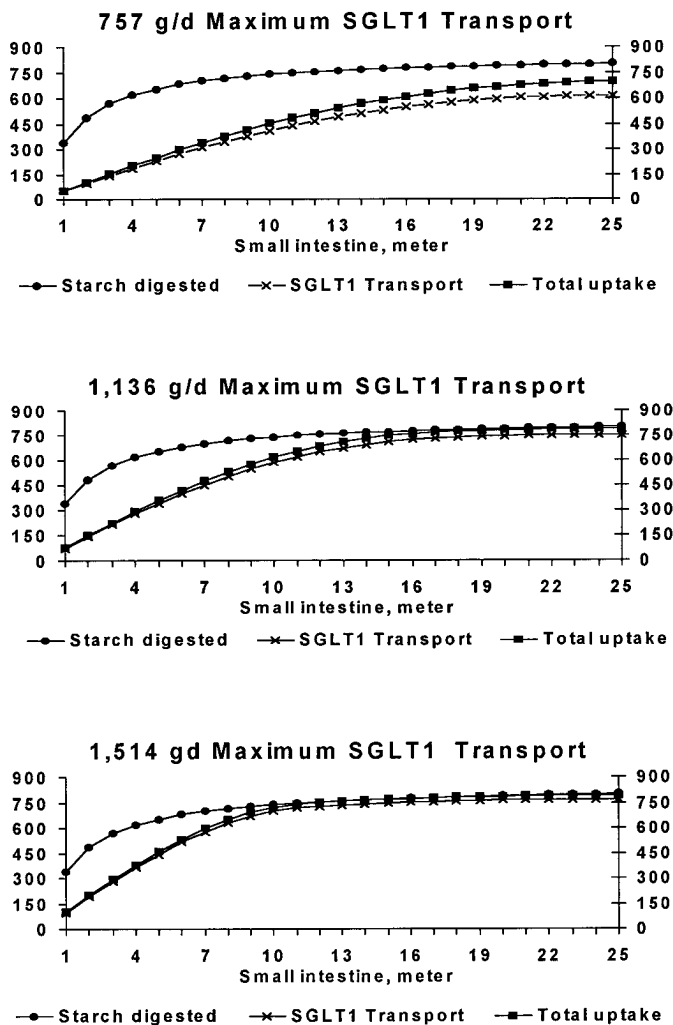


Figure 3. Simulations of cumulative starch digestion, glucose transport by SGLT1, and total transport (SGLT1 plus paracellular diffusion) at three levels of maximal SGLT1 transport. Starch digestion is the same in all three simulations, and matches the highest observed digestion reported by Kreikemeier et al. (1991).

glucose; therefore, the simulation seems to reasonably represent a practical, feedlot situation. The simulation does not leave much opportunity for intestinal fermentation of glucose, because luminal concentrations in the intestine of adapted steers rapidly approach zero (Figure 2).

The simulation was extended to a 350-kg beef steer eating a high concentrate diet and to a 520-kg lactating dairy cow; those examples are described in Table 3. Intestinal length, volume, water absorption, and paracellular diffusion for the steer were the same as those used to develop the simulation. Maximal SGLT1 transport capacity was set at 1,514 g/d and K_m at 5 mM. Duodenal starch entry was 2,058 g/d. Intestinal length for the dairy cow was set at 30 m, ileal flow (73 L/d) was from Lu et al. (1988), duodenal starch entry was set at 2,790 g/d, and maximal SGLT1 transport capacity was set at 2,725 g/

d (the basal value for the steer from the simulation [757 g/d] multiplied by 1.5 to arbitrarily adjust for body size and metabolic activity, 1.2 to accommodate a longer intestine, and 2.0 to adapt to starch flow to the duodenum). Intestinal volume (6.62 L) was calculated from fluid loss (127 L/d) and the principle of transport of .8 times intestinal luminal water volume each hour (Pappenheimer and Reiss, 1987). Paracellular diffusion used the same equations that were used to develop the simulation. The condensed version of glucose uptake by diffusion was diffusion (grams/day) equals 1.356 times millimolar luminal glucose concentration.

Because the amount of starch digested was appreciably less than maximal SGLT1 transport capacity, both the steer and lactating cow were able to transport almost all glucose available from starch digestion in the proximal half of the intestine (Figure 4). One can conclude from these simulations that in adapted animals, capacity to digest starch is the primary limitation to maximal capture of glucose from intestinal starch digestion until or unless that capacity exceeds approximately 1.5 kg/d for steers and 3.0 kg/d for cows.

Insufficient Digestive Enzymes as the Primary Limitation to Intestinal Starch Digestion

The review of Owens et al. (1986), the simulations described above, and the results of Kreikemeier et al. (1991) suggest that amylase likely is more limiting than intestinal oligosaccharidase activity in the starch

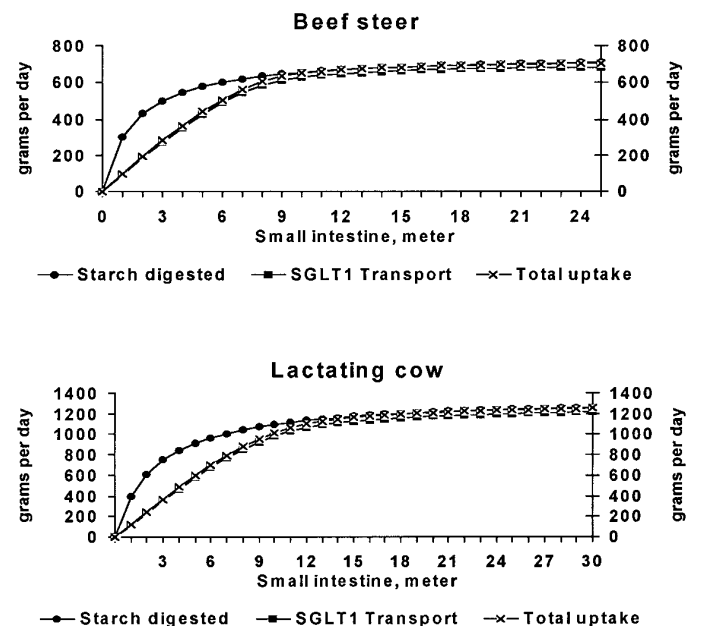


Figure 4. Simulations of cumulative starch digestion, glucose transport by SGLT1, and total transport (SGLT1 plus paracellular diffusion) for the beef steer and the dairy cow described in Table 3.

digestion process. Therefore, increased effective supply of amylase should enhance intestinal starch digestibility and absorption as glucose. Efforts to increase pancreatic output of digestive enzymes by injection of slaframine were not successful. The pancreas of steers released more fluid in response to the drug, but the fluid had lower concentration of enzymes. Therefore, increased postruminal digestion of starch in steers injected with slaframine (Walker et al., 1994) did not seem to be due to increased supply of pancreatic amylase. Recent work with sheep and steers lends support to the established idea that more protein available for digestion in the small intestine means greater secretion of all digestive enzymes for the pancreas, including those responsible for starch digestion. Taniguchi et al. (1993, 1995) showed increased digestibility of starch in the small intestine of sheep and increased net appearance of glucose in portal blood of steers in direct response to increased supply of protein to the small intestine. These data and others already discussed indicate a double benefit from ruminal digestion of starch: first, increased production and outflow of microbial protein from the rumen; and second, increased duodenal digestion of starch as a result of the pancreatic response to more protein supply entering the small intestine.

Glucose Metabolism

Metabolic needs or uses for glucose include certain functions of the nervous system, generation of ATP through glycolysis and the tricarboxylic acid cycle, and generation of reducing equivalents (NADPH) through the hexose monophosphate shunt. Changes in nutritional or physiological state (fasting, rapid growth, pregnancy, lactation) affect amounts and rates of glucose use. Glucose metabolism is closely associated with metabolism of amino acids and lipids through the endocrine action of insulin and glucagon. Blood concentrations of glucose, urea, amino acids, NEFA, ketones, and VFA usually decrease in response to elevated concentration of insulin (Bell et al., 1987; Debras et al., 1988; Faulkner and Pollock, 1990; Petterson et al., 1993; Eisemann and Huntington, 1994). Increased plasma insulin concentration is linked to decreased hepatic gluconeogenesis and to increased glucose use by peripheral tissues except the mammary gland (Sano et al., 1992; Eisemann and Huntington, 1994; McGuire et al., 1995). Removal of propionate (a major glucose precursor) by the liver in fed steers was not greatly affected by insulin (Eisemann and Huntington, 1994). Therefore, decreased glucose concentrations in blood are due to the combined effects of decreased production and increased use, just as changes in concentrations of insulin or glucagon may be due to changes in production of those hormones by pancreas and(or)

removal by target tissues (Guerino et al., 1991). Glucagon generally has countervailing effects to those of insulin, but concentrations of both may rise or fall together (Brockman, 1978).

Ruminants eating high-forage, low-starch diets depend on liver synthesis of glucose to meet their metabolic requirements. Principal substrates or carbon sources for glucose synthesis are organic acids from fermentation (mainly propionate and lactate), the carbon skeletons of deaminated amino acids, and glycerol from breakdown of triglycerides. The ruminant's liver removes essentially all three-, four-, and five-carbon volatile fatty acids that arrive in portal blood draining the gut. The liver also removes substantial portions of lactate and amino acids that arrive in portal blood (Huntington, 1990). Because glucose use increases as energy intake increases (Herbein et al., 1978; Schmidt and Keith, 1983), ruminants' glucose requirement as well as potential dietary sources of glucose increase when they eat high-starch diets. As stated previously, pancreatic secretion of digestive enzymes likewise increases with increased energy intake. With high-starch diets, glucose synthesis in the liver follows the general scheme outlined for high-forage diets, except for possible changes in relative contribution of the various sources of carbon that are used to synthesize glucose. Results of studies with multicatheterized ruminants indicate that net release of glucose by the liver may (Taniguchi et al., 1995) or may not change (Reynolds and Tyrrell, 1991; Casse et al., 1994) with increased supply of glucose or glucose precursors from the portal-drained viscera, but proportional use of precursors by the liver does reflect changes in that supply (Reynolds and Tyrrell, 1991; Casse et al., 1994; Bauer et al., 1995; Taniguchi et al., 1995). Studies with isotopes have shown that propionate from ruminal fermentation provides from 43 to 67% of carbon used for glucose synthesis in the liver and that lactate provides up to 12% (Huntington et al., 1981; Veenhuizen et al., 1988; Amaral et al., 1990).

Table 3 contains two examples to show how starch digestion and glucose absorption relate to overall glucose metabolism in a beef steer eating a diet containing 77% dry-rolled sorghum (Theurer et al., 1990, 1991) and in a first-lactation dairy cow eating a diet containing 47% ground corn and 3% ground barley (Casse et al., 1994). Intestinal starch digestion and glucose uptake from the intestine for these examples are in Figure 4. The researchers measured net absorption of glucose and other nutrients from the gut, liver uptake of glucose precursors, liver output of glucose, milk production of the cows, and starch and digestible energy intake of the steers. Starch intake of the cow was estimated from average starch concentrations in feedstuffs cited earlier in this review. Digestion of starch ruminally and postruminally for both the steer and the cow was calculated from values in

Table 3. Starch digestion and glucose kinetics in a beef steer and a lactating dairy cow

Item	Beef steer ^a	Dairy cow ^b
Body weight, kg	363	520
Weight gain/milk yield, kg/d	1.1	32
Digestible energy intake, Mcal/d	21.8	44.0
Starch source	Dry-rolled sorghum	Ground corn and barley
Starch intake, g/d ^c	4,278	5,690
Ruminal starch digestion, g/d ^d	2,220	2,900
Postruminal starch digestion, g/d ^d	1,286	2,320
Glucose variable		
Irreversible loss, g/d ^e	1,096	2,007
Oxidized to CO ₂ , g/d ^f	487	345
Secreted as milk lactose, g/d ^g	0	1,600
Net liver glucose output, g/d	1,443	3,313
Net output by the viscera, g/d	-60	220
Glucose from starch digestion, g/d ^h	707	1,276
Total glucose entry, g/d ⁱ	2,150	4,589
Maximal contribution to liver glucose, %		
Propionate	58	88
Lactate	7	6

^aFrom data of Theurer et al. (1990, 1991).

^bFrom data of Casse et al. (1992).

^cMeasured for steers, calculated for cows.

^dCalculated from digestibilities in Table 1.

^eIrreversible loss (mol/d) = $1.116 + .228 \text{ DE (Mcal/d)}$ (Herbein et al., 1978). Represents glucose that leaves blood plasma never to return.

^fIrreversible loss $\times .444$ for steers (Veenhuizen et al., 1988) and irreversible loss $\times .172$ for cows (Bauman et al., 1988).

^gCalculated as measured milk yield $\times .05$.

^hCalculated as postruminal starch digested $\times .55$ (Kreikemeier et al., 1991).

ⁱSum of net liver production and glucose from starch digestion. Represents all new glucose that enters the blood plasma.

Table 1. Sources of other factors used in the calculations are listed in the footnotes of Table 3. Irreversible loss represents glucose that leaves blood plasma never to return as glucose; it can be regarded as the animal's minimal requirement for new glucose. Total glucose entry is the amount of new glucose that enters blood plasma from outside sources (absorption from the small intestine and synthesis by the liver). The difference between total entry and irreversible loss represents the amount of glucose that is recycled (leaves blood plasma to participate in metabolism but reenters as glucose). The equation for calculating glucose irreversible loss from digestible energy intake (Herbein et al., 1978) is reliable, based on agreement of that equation's prediction with independent data sets (Weighart et al., 1986; Bauman et al., 1988). As discussed previously, glucose absorption could be calculated as .45 to .88 times postruminal starch digested; .55 was used in Table 3. This factor has a big effect on the calculation of total glucose entry. Comparison of the proportion of total glucose entry accounted for by irreversible loss to the data of Schmidt and Keith (1983) indicates that total glucose entry may be overestimated but is not underestimated in Table 3.

Compared with the steer, the cow in Table 3 eats about twice as much digestible energy, has twice the

postruminal starch digestion and glucose absorption, twice the irreversible loss of glucose, and twice the total glucose entry. Net liver output of glucose for the cow is greater than twice that for the steer. For both the steer and cow, net liver output exceeds irreversible loss, which exemplifies the ruminants' ability to meet their minimal glucose needs by liver synthesis, or their independence from exogenous glucose supply. Starch intake and ruminal digestion favor the cow as well, but not by a factor of two. Differences in postruminal starch digestion reflect the high ruminal outflow rates for dairy cows (McCarthy et al., 1989), which counterbalance differences in ruminal digestibility of corn and barley vs sorghum.

For both the steer and cow in Table 3, glucose absorbed provides similar percentages of total glucose entry (28 to 33%) and of irreversible loss (64 to 65%), and the steer and cow have similar maximal theoretical contribution of lactate to liver glucose synthesis (6 to 7%). Maximal theoretical contribution of propionate to glucose synthesis in the liver is higher for the cow; propionate maximally can account for 64% of total glucose entry whereas the steer gets up to 39% of total glucose entry from propionate uptake by the liver. These percentages for lactate and propionate are maximal theoretical, because they are based on the presumption that all liver removal of those precursors

leaves the liver in the form of glucose. Ostensibly, other sources of glucose carbon, such as glycerol from lipid turnover or carbon from amino acids, are more important for the beef steer. The examples in Table 3 show how production needs affect the partition of glucose for use by visceral tissues, and they show how growth or lactation affects the amounts of glucose that are recycled, oxidized to CO_2 , or used by the mammary gland. Net glucose output by the viscera is negative for the steer and positive for the dairy cow. That means that the viscera (gut, pancreas, spleen, omentum, and mesentery) of the steer uses 60 g of glucose/d more than it absorbs. Total visceral use by the steer is 767 g/d or 36% of total glucose entry. The viscera of the cow uses 220 g less than is absorbed, 1,056 g/d or 23% of glucose entry. The glucose excreted as milk lactose by the dairy cow is about 1.5 times the whole-body irreversible loss of the steer. Glucose recycled (the difference between entry rate and irreversible loss) is 2.4 times larger for the dairy cow than for the steer. However, the steer oxidizes to CO_2 1.4 times more glucose than the dairy cow. Compared with the steer, the dairy cow obtains similar proportions of glucose supply from the diet, but she is recycling more glucose and oxidizing less glucose so she can meet her need to synthesize and excrete milk lactose.

A final point to be made from Table 3 is the relation between starch intake and glucose metabolism. Although no example fits all production situations, these examples are representative of the "middle ground" of beef or milk production in the United States. Total glucose entry accounts for 50% of starch intake of the steer and 80% of starch intake of the cow. Some losses of starch intake, such as incomplete digestion and absorption, fermentation to methane, and oxidation to CO_2 , will not be avoided in profitable production situations. However, the examples show that the steer has a more than adequate supply of glucose from dietary sources and suggest that ruminal fermentation of the starch to provide other nutrient needs (energy substrates, amino acids) would improve the efficiency of starch use for growth; results previously discussed and those to be discussed in this review imply that this is the case. On the other hand, the dairy cow is much closer to optimal use of starch to supply her glucose needs; the glucose supply from starch digestion is close to her ability to actively transport glucose across the gut wall, and she is recycling 7.5 times more glucose than she oxidizes to CO_2 . Both the steer and the cow will benefit more from a supply of readily fermented starch (to augment ruminal production of nutrients) than from an increased supply of starch that escapes ruminal fermentation.

Summary and Conclusions

All grains except oats respond positively to processing, particularly processing that involves application

of steam and mechanical rolling or flaking. The response is most dramatic for sorghum and least dramatic for wheat. Differences among sorghum hybrids are more a function of the type and location of protein than total protein content. The advent and application of new laboratory techniques have enhanced our understanding of the taxonomy and function of ruminal microbes. Further, application of new analytical techniques provides rapid and fairly accurate assessment of sites and extents of digestion in the ruminant's digestive tract, which improves the ability to evaluate a feed source and accurately predict performance of animals that consume that feed. The challenge for the production manager is to adjust processing techniques and diet formulation to accommodate the characteristics of the available sources of starch.

Simulations of starch digestion and glucose uptake in the intestine of a growing steer and a lactating cow suggest that capacity to digest starch is the primary factor that limits absorption of glucose in the small intestine. At production levels of intake in animals adapted to high-starch diets, simulated glucose transport by the SGLT1 transporter exceeds putative enzymatic starch digestion by pancreatic amylases and intestinal oligosaccharidases almost by a factor of two. Glucose uptake by paracellular diffusion is a minor contributor to total glucose uptake in the small intestine, because simulated luminal glucose concentrations approach only 30 mM, and then for a limited period or space in the intestine.

Starch use by ruminants should be related to the animal's glucose needs. My examples show that glucose absorption contributes 33% of the total glucose supply of a beef steer; 44% of that glucose supply comes from organic acids from starch fermentation in the rumen and 23% from other carbon sources, such as amino acids. For a lactating cow, 28% of total glucose supply comes from absorbed glucose, 67% from organic acids from starch fermentation in the rumen, and 5% from other sources. As a general rule, the amount of glucose used by visceral tissues of ruminants eating high grain diets is equal to or slightly greater than the amount of glucose absorbed from the small intestine.

Implications

Intensive studies that have quantified the limits on glucose supply from postruminal digestion, and production studies with processed grains indicate decisively that grain starch is used best when it is extensively fermented in the rumen. The principal responses of grains to processing are decreased density and increased fermentation in the rumen. Because digestive capacity limits maximal capture of starch entering the small intestine, approximately 45% of starch entering the intestine is not absorbed as glucose. Therefore, any improvement in metabolic

energetics attributable to increased glucose supply from dietary sources must consider potential losses of energy attributable to fermentation of starch in the cecum, large intestine, and colon rather than in the rumen. The choice of an optimal level of dietary starch and site of starch digestion from an economic standpoint will match starch digestion to the glucose needs of the animal under a given production scenario.

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